

MIPS Reference Data

①



CORE INSTRUCTION SET

NAME, MNEMONIC	FOR-MAT	OPERATION (in Verilog)	OPCODE / FUNCT (Hex)
Add	add R	$R[rd] = R[rs] + R[rt]$	(1) $0 / 20_{hex}$
Add Immediate	addi I	$R[rt] = R[rs] + \text{SignExtImm}$	(1,2) 8_{hex}
Add Imm. Unsigned	addiu I	$R[rt] = R[rs] + \text{SignExtImm}$	(2) 9_{hex}
Add Unsigned	addu R	$R[rd] = R[rs] + R[rt]$	$0 / 21_{hex}$
And	and R	$R[rd] = R[rs] \& R[rt]$	$0 / 24_{hex}$
And Immediate	andi I	$R[rt] = R[rs] \& \text{ZeroExtImm}$	(3) C_{hex}
Branch On Equal	beq I	if $R[rs] == R[rt]$ $PC = PC + 4 + \text{BranchAddr}$	(4) 4_{hex}
Branch On Not Equal	bne I	if $R[rs] != R[rt]$ $PC = PC + 4 + \text{BranchAddr}$	(4) 5_{hex}
Jump	j J	$PC = \text{JumpAddr}$	(5) 2_{hex}
Jump And Link	jal J	$R[31] = PC + 8; PC = \text{JumpAddr}$	(5) 3_{hex}
Jump Register	jr R	$PC = R[rs]$	$0 / 08_{hex}$
Load Byte Unsigned	lbu I	$R[rt] = \{24'b0, M[R[rs]] + \text{SignExtImm}(7:0)\}$	(2) 24_{hex}
Load Halfword Unsigned	lhu I	$R[rt] = \{16'b0, M[R[rs]] + \text{SignExtImm}(15:0)\}$	(2) 25_{hex}
Load Linked	ll I	$R[rt] = M[R[rs] + \text{SignExtImm}]$	(2,7) 30_{hex}
Load Upper Imm.	lui I	$R[rt] = \{\text{imm}, 16'b0\}$	f_{hex}
Load Word	lw I	$R[rt] = M[R[rs] + \text{SignExtImm}]$	(2) 23_{hex}
Nor	nor R	$R[rd] = \sim (R[rs] R[rt])$	$0 / 27_{hex}$
Or	or R	$R[rd] = R[rs] R[rt]$	$0 / 25_{hex}$
Or Immediate	ori I	$R[rt] = R[rs] \text{ZeroExtImm}$	(3) d_{hex}
Set Less Than	slt R	$R[rd] = (R[rs] < R[rt]) ? 1 : 0$	$0 / 2a_{hex}$
Set Less Than Imm.	slti I	$R[rt] = (R[rs] < \text{SignExtImm}) ? 1 : 0$	(2) a_{hex}
Set Less Than Imm. Unsigned	sltiu I	$R[rt] = (R[rs] < \text{SignExtImm}) ? 1 : 0$	(2,6) b_{hex}
Set Less Than Unsig.	sltu R	$R[rd] = (R[rs] < R[rt]) ? 1 : 0$	(6) $0 / 2b_{hex}$
Shift Left Logical	sll R	$R[rd] = R[rt] \ll \text{shamt}$	$0 / 00_{hex}$
Shift Right Logical	srl R	$R[rd] = R[rt] \gg \text{shamt}$	$0 / 02_{hex}$
Store Byte	sb I	$M[R[rs] + \text{SignExtImm}](7:0) = R[rt](7:0)$	(2) 28_{hex}
Store Conditional	sc I	$M[R[rs] + \text{SignExtImm}] = R[rt];$ $R[rt] = (\text{atomic}) ? 1 : 0$	(2,7) 38_{hex}
Store Halfword	sh I	$M[R[rs] + \text{SignExtImm}](15:0) = R[rt](15:0)$	(2) 29_{hex}
Store Word	sw I	$M[R[rs] + \text{SignExtImm}] = R[rt]$	(2) $2b_{hex}$
Subtract	sub R	$R[rd] = R[rs] - R[rt]$	(1) $0 / 22_{hex}$
Subtract Unsigned	subu R	$R[rd] = R[rs] - R[rt]$	$0 / 23_{hex}$

- (1) May cause overflow exception
- (2) $\text{SignExtImm} = \{16\{\text{immediate}[15]\}, \text{immediate}\}$
- (3) $\text{ZeroExtImm} = \{16\{1'b0\}, \text{immediate}\}$
- (4) $\text{BranchAddr} = \{14\{\text{immediate}[15]\}, \text{immediate}, 2'b0\}$
- (5) $\text{JumpAddr} = \{PC + 4[31:28], \text{address}, 2'b0\}$
- (6) Operands considered unsigned numbers (vs. 2's comp.)
- (7) Atomic test&set pair; $R[rt] = 1$ if pair atomic, 0 if not atomic

BASIC INSTRUCTION FORMATS

R	opcode		rs	rt		rd	shamt	funct
	31	26 25	21 20	16 15		11 10	6 5	0
I	opcode		rs	rt	immediate			
	31	26 25	21 20	16 15		0		
J	opcode		address					
	31	26 25	0					

ARITHMETIC CORE INSTRUCTION SET

NAME, MNEMONIC	FOR-MAT	OPERATION	OPCODE / FUNCT (Hex)
Branch On FP True	bclt FI	if $(FPcond) PC = PC + 4 + \text{BranchAddr}$	(4) $11/8/--$
Branch On FP False	bclt FI	if $(!FPcond) PC = PC + 4 + \text{BranchAddr}$	(4) $11/8/0/--$
Divide	div R	$Lo = R[rs]/R[rt]; Hi = R[rs]\%R[rt]$	$0/--/--1a$
Divide Unsigned	divu R	$Lo = R[rs]/R[rt]; Hi = R[rs]\%R[rt]$	(6) $0/--/--1b$
FP Add Single	add.s FR	$F[fd] = F[fs] + F[ft]$	$11/10/--/0$
FP Add Double	add.d FR	$\{F[fd], F[fd+1]\} = \{F[fs], F[fs+1]\} + \{F[ft], F[ft+1]\}$	$11/11/--/0$
FP Compare Single	c.x.s* FR	$FPcond = (F[fs] op F[ft]) ? 1 : 0$	$11/10/--/y$
FP Compare Double	c.x.d* FR	$FPcond = (\{F[fs], F[fs+1]\} op \{F[ft], F[ft+1]\}) ? 1 : 0$	$11/11/--/y$
* (x is eq, lt, or le) (op is ==, <, or <=) (y is 32, 3c, or 3e)			
FP Divide Single	div.s FR	$F[fd] = F[fs] / F[ft]$	$11/10/--/3$
FP Divide Double	div.d FR	$\{F[fd], F[fd+1]\} = \{F[fs], F[fs+1]\} / \{F[ft], F[ft+1]\}$	$11/11/--/3$
FP Multiply Single	mul.s FR	$F[fd] = F[fs] * F[ft]$	$11/10/--/2$
FP Multiply Double	mul.d FR	$\{F[fd], F[fd+1]\} = \{F[fs], F[fs+1]\} * \{F[ft], F[ft+1]\}$	$11/11/--/2$
FP Subtract Single	sub.s FR	$F[fd] = F[fs] - F[ft]$	$11/10/--/1$
FP Subtract Double	sub.d FR	$\{F[fd], F[fd+1]\} = \{F[fs], F[fs+1]\} - \{F[ft], F[ft+1]\}$	$11/11/--/1$
Load FP Single	lwc1 I	$F[rt] = M[R[rs] + \text{SignExtImm}]$	(2) $31/--/--$
Load FP Double	ldc1 I	$F[rt] = M[R[rs] + \text{SignExtImm}];$ $F[rt+1] = M[R[rs] + \text{SignExtImm} + 4]$	(2) $35/--/--$
Move From Hi	mfhi R	$R[rd] = Hi$	$0/--/--10$
Move From Lo	mflo R	$R[rd] = Lo$	$0/--/--12$
Move From Control	mfc0 R	$R[rd] = CR[rs]$	$10/0/--0$
Multiply	mult R	$\{Hi, Lo\} = R[rs] * R[rt]$	$0/--/--18$
Multiply Unsigned	multu R	$\{Hi, Lo\} = R[rs] * R[rt]$	(6) $0/--/--19$
Shift Right Arith.	sra R	$R[rd] = R[rt] \gg \text{shamt}$	$0/--/--3$
Store FP Single	swc1 I	$M[R[rs] + \text{SignExtImm}] = F[rt]$	(2) $39/--/--$
Store FP Double	sdc1 I	$M[R[rs] + \text{SignExtImm}] = F[rt];$ $M[R[rs] + \text{SignExtImm} + 4] = F[rt+1]$	(2) $3d/--/--$

FLOATING-POINT INSTRUCTION FORMATS

FR	opcode	fmt	ft	fs	fd	funct
	31	26 25	21 20	16 15	11 10	6 5
FI	opcode	fmt	ft	immediate		
	31	26 25	21 20	16 15		

PSEUDOINSTRUCTION SET

NAME	MNEMONIC	OPERATION
Branch Less Than	blt	if $R[rs] < R[rt]$ $PC = \text{Label}$
Branch Greater Than	bgt	if $R[rs] > R[rt]$ $PC = \text{Label}$
Branch Less Than or Equal	bte	if $R[rs] \leq R[rt]$ $PC = \text{Label}$
Branch Greater Than or Equal	bge	if $R[rs] \geq R[rt]$ $PC = \text{Label}$
Load Immediate	li	$R[rd] = \text{immediate}$
Move	move	$R[rd] = R[rs]$

REGISTER NAME, NUMBER, USE, CALL CONVENTION

NAME	NUMBER	USE	PRESERVED ACROSS A CALL?
\$zero	0	The Constant Value 0	N.A.
\$at	1	Assembler Temporary	No
\$v0-\$v1	2-3	Values for Function Results and Expression Evaluation	No
\$a0-\$a3	4-7	Arguments	No
\$t0-\$t7	8-15	Temporaries	No
\$s0-\$s7	16-23	Saved Temporaries	Yes
\$t8-\$t9	24-25	Temporaries	No
\$k0-\$k1	26-27	Reserved for OS Kernel	No
\$gp	28	Global Pointer	Yes
\$sp	29	Stack Pointer	Yes
\$fp	30	Frame Pointer	Yes
\$ra	31	Return Address	Yes